

**BENHA NATIONAL UNIVERSITY
MECHATRONICS ENGINEERING**

**6DOF ROBOT ARM
INVERSE KINEMATICS REPORT**

1. Input Parameters

Parameter	Value
X (mm)	180
Y (mm)	50
Z (mm)	180
Roll	0
Pitch	0
Yaw	0
a2	160.0
a3	140.0

2. Joint Angles (IK Result)

Joint	Angle (deg)
θ_1	0.000
θ_2	0.000
θ_3	0.000
θ_4	0.000
θ_5	0.000
θ_6	0.000

3. Workspace Analysis

Wrist Center X	0.00
Wrist Center Y	0.00
Wrist Center Z	0.00
r	0.00
s	0.00
D	0.0000
Reach	0.00
Max Reach	0.00
Status	INSIDE WORKSPACE

4. System Status

Status: WAITING...

5. DH Transformation Matrices

6. Final Transformation Matrix (T06)